# Exercise Sheet 6

## Evaluation and Empirical Risk Minimisation

Probabilistic Machine Learning

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# 6.A Measuring a Rare Class

# Exercise 6.A.1: A New Fraud Model

A new fraud model is tested on **200,000** transactions, one in 1,000 of them fraudulent. It flags **750** transactions, **150** of which are frauds.

1. Fill in the confusion matrix (TP, FN, FP, TN).
2. Compute the model's accuracy, recall and precision, and the accuracy of the "always legitimate" baseline.
3. A check costs €5, a missed fraud €500. Compute the total cost of the model, of "always legitimate" and of "flag everything".
4. The model is moved to a market where one transaction in **100** is fraudulent. Its recall and its false-alarm rate stay the same. What is its precision there?

# Exercise 6.A.1: A New Fraud Model — Solution

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**1.** 200 frauds, 199,800 legitimate: TP = 150, FN = 50, FP = 600, TN = 199,200.

**2.** Accuracy $= (150 + 199{,}200)/200{,}000 = 0.99675$; baseline $199{,}800/200{,}000 = 0.999$ — the baseline "wins" again. Recall $= 150/200 = 0.75$, precision $= 150/750 = 0.20$.

**3.** Model: 50 · €500 + 600 · €5 = €28,000. Always legitimate: 200 · €500 = €100,000. Flag everything: 199,800 · €5 = €999,000. The model is by far the cheapest although its accuracy is the lowest of the two sensible options.

**4.** False-alarm rate $600/199{,}800 \approx 0.3\%$. In 200,000 transactions at 1/100: 2,000 frauds, TP $= 0.75 \cdot 2{,}000 = 1{,}500$, FP $\approx 0.003 \cdot 198{,}000 \approx 595$. Precision $= 1{,}500/2{,}095 \approx 0.72$. Precision depends on the base rate, recall does not.

# Exercise 6.A.2: The Bayes Error

A single attribute $x$ takes three values:

| $x$ | $P(x)$ | $P(y = 1 \mid x)$ |
|---|---|---|
| A | 0.5 | 0.1 |
| B | 0.3 | 0.5 |
| C | 0.2 | 0.8 |

1. Which classifier has the smallest true error (0/1 loss)? Compute its error — the Bayes error.
2. What is the error of "always predict 0"?
3. A team reports a test error of **0.20** on **100** test cases. Is that evidence that their model beats the Bayes error? What if the same 0.20 came from **10,000** test cases?
4. Would more rows of the same data lower the Bayes error? What would?

# Exercise 6.A.2: The Bayes Error — Solution

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**1.** Predict the more probable label for each $x$: 0 for A, 1 for C, either for B. Error $= 0.5 \cdot 0.1 + 0.3 \cdot 0.5 + 0.2 \cdot 0.2 = 0.24$.

**2.** $P(y = 1) = 0.05 + 0.15 + 0.16 = 0.36$.

**3.** $\mathrm{SE} = \sqrt{0.2 \cdot 0.8/100} = 0.04$, so the test error is $0.20 \pm 0.08$ (±2 SE) — compatible with 0.24; no evidence of anything. On 10,000 cases, $\mathrm{SE} = 0.004$ and $0.20 \pm 0.008$ excludes 0.24: no classifier can do that in expectation, so look for a **leak** (test cases seen in training, a feature that encodes the label) or a test set not drawn from the same distribution.

**4.** No — the 0.24 is aleatoric relative to $x$. Only a **new attribute** that separates the cases inside B (and A, C) can lower it.

# 6.B Numbers With Error Bars

# Exercise 6.B.1: Error Bars on a Recall

On a test set of **50,000** transactions there are **50** frauds; model A catches **40** of them.

1. Compute the recall, its standard error and a ±2 SE interval.
2. With a flat prior, what is the Beta posterior for the recall? Give its mean. (Its 95% credible interval is $[0.67, 0.89]$.)
3. How many test frauds — and roughly how many transactions — are needed for an error bar of ±0.02 (±2 SE) at a recall near 0.8?
4. Model B catches 42 of the 50. The two models disagree on **5** frauds: B caught 4 of them, A caught 1. If both models were equally good, each disagreement would go either way with probability ½. How likely is a split at least as lopsided as 4 : 1 in B's favour? Conclusion?

# Exercise 6.B.1: Error Bars on a Recall — Solution

# Exercise 6.B.1: Error Bars on a Recall — Solution

**1.** Recall $= 40/50 = 0.80$, $\mathrm{SE} = \sqrt{0.8 \cdot 0.2/50} \approx 0.057$, interval $\approx [0.69, 0.91]$.

**2.** $\mathrm{Beta}(1 + 40, 1 + 10) = \mathrm{Beta}(41, 11)$, mean $41/52 \approx 0.79$ — pulled slightly towards 0.5 by the flat prior's one pseudo-catch and one pseudo-miss.

**3.** $\mathrm{SE} = 0.01 : 0.16/n = 0.0001$, so $n = 1{,}600$ frauds, about **1.6 million** transactions at one in 1,000.

**4.** $P(4 \text{ or } 5 \text{ of } 5) = (5 + 1)/32 \approx 0.19$. A split this lopsided happens about one time in five by chance: no evidence that B is better. Only the five disagreements carry information about the difference — the 45 frauds that both caught or both missed do not.

# 6.C From Scores to Decisions

# Exercise 6.C.1: Thresholds From Costs

A check costs €5 ($c_{FP}$), a missed fraud €500 ($c_{FN}$).

1. Derive the threshold $p^*$ above which checking is cheaper in expectation.
2. The bank buys chargeback insurance, and a missed fraud now costs **€100**. What is the new threshold?
3. Four transactions have calibrated scores 0.002, 0.03, 0.09 and 0.4. Which are checked under the old and under the new costs? Verify the 0.03 case with expected costs.
4. It turns out the model's scores are **twice** the true probabilities. What goes wrong with the rule of part 1, and how do you repair it?

# Exercise 6.C.1: Thresholds From Costs — Solution

# Exercise 6.C.1: Thresholds From Costs — Solution

**1.** Check iff $(1-p)\cdot c_{FP} < p \cdot c_{FN}$, i.e. $p > p^* = c_{FP}/(c_{FP}+c_{FN}) = 5/505 \approx 0.0099$.

**2.** $p^* = 5/105 \approx 0.048$.

**3.** Old costs: check 0.03, 0.09, 0.4; let 0.002 through. New costs: check 0.09 and 0.4 only. For 0.03 under the new costs: letting through costs 0.03 · €100 = €3, checking 0.97 · €5 = €4.85 — let it through. Under the old costs, letting through would cost €15.

**4.** A score of 0.0099 means a true probability of about 0.005, so the rule checks transactions whose true $p$ is only half the break-even: too many checks. Either calibrate the scores (Lecture 04: reliability diagram on held-out data, then refit) or, equivalently here, apply the threshold $2p^* \approx 0.02$ to the raw score. The cost rule needs **calibrated** probabilities.

# Exercise 6.C.2: Scoring Probabilities

Five held-out transactions with outcomes $y = (1, 0, 0, 0, 1)$ (1 = fraud).

| | $p_1$ | $p_2$ | $p_3$ | $p_4$ | $p_5$ |
|---|---|---|---|---|---|
| model A | 0.8 | 0.1 | 0.2 | 0.1 | 0.6 |
| model B | 0.99 | 0.01 | 0.01 | 0.01 | 0.4 |

1. Compute each model's accuracy at the threshold 0.5.
2. Compute each model's log-loss (natural logarithm) and Brier score.
3. The three measures do not agree. What does each one reward?

# Exercise 6.C.2: Scoring Probabilities — Solution

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**1.** A: 5/5 = 1.0. B misses the last fraud ($0.4 < 0.5$): 4/5 = 0.8.

**2.** Log-loss, A: $(0.223 + 0.105 + 0.223 + 0.105 + 0.511)/5 \approx 0.234$. B: $(4 \cdot 0.010 + 0.916)/5 \approx 0.191$. Brier, A: $(0.04 + 0.01 + 0.04 + 0.01 + 0.16)/5 = 0.052$; B: $(4 \cdot 0.0001 + 0.36)/5 \approx 0.072$.

**3.** Accuracy rewards being on the right side of 0.5 and ignores how far — it prefers A. Log-loss rewards confident correct predictions strongly and is lenient to B's single moderate miss (0.4 on a fraud is not a confident error), so it prefers B. Brier is bounded and closer to a squared distance, so the one 0.4 miss weighs more and it prefers A. Both log-loss and Brier are proper; with five cases, neither difference is meaningful — the error-bar lesson of 6.B applies to these scores too.

# 6.D The Shortcut: Keep the Loss

# Exercise 6.D.1: What Is ERM, and What Does It Drop?

1. Show that the MAP estimate for linear regression with Gaussian noise $\mathcal{N}(0, \sigma^2)$ and prior $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \tau^2\mathbf{I})$ minimises $\frac{1}{n}\sum_i (y_i - \mathbf{w}^\top \mathbf{x}_i)^2 + \lambda \|\mathbf{w}\|^2$. Express $\lambda$ in $\sigma, \tau$ and $n$.
2. Evaluate $\lambda$ for $\sigma = 40, \tau = 10$ and $n = 20$, then $n = 2{,}000$. What happens to the prior?
3. For each method, say whether it is ERM in the form $\min_h \frac{1}{n}\sum_i L(h(\mathbf{x}_i), y_i)$, and whether its output is a probability: logistic regression · linear SVM · naive Bayes (maximum likelihood) · Bayesian linear regression.

# Exercise 6.D.1: What Is ERM, and What Does It Drop? — Solution

# Exercise 6.D.1: What Is ERM, and What Does It Drop? — Solution

**1.** $-\log p(\mathbf{w} \mid D) = \frac{1}{2\sigma^2}\sum_i (y_i - \mathbf{w}^\top \mathbf{x}_i)^2 + \frac{1}{2\tau^2}\|\mathbf{w}\|^2 + \text{const}$. Multiplying by $2\sigma^2/n$ does not move the minimum and gives the stated form with $\lambda = \sigma^2/(n\tau^2)$.

**2.** $n = 20$: $\lambda = 1{,}600/2{,}000 = 0.8$. $n = 2{,}000$: $\lambda = 0.008$. The prior's weight falls like $1/n$ — Lecture 04's pseudo-counts being swamped by data.

**3.**

| method | ERM form? | output a probability? |
|---|---|---|
| logistic regression | yes, $L = -\log p(y \mid \mathbf{x})$ | yes |
| linear SVM | yes, hinge loss (+ penalty) | **no** — a margin score |
| naive Bayes | no — maximises $\log p(\mathbf{x}, y)$ | yes (often poorly calibrated) |
| Bayesian linear regression | no — nothing is minimised | yes, a whole predictive distribution |